or data networks. Cells will often be smaller in size in large towns, as the number of users in the area is more. Communication over mobile network is made up of voice, data, images and text messages. See Figure 11.5

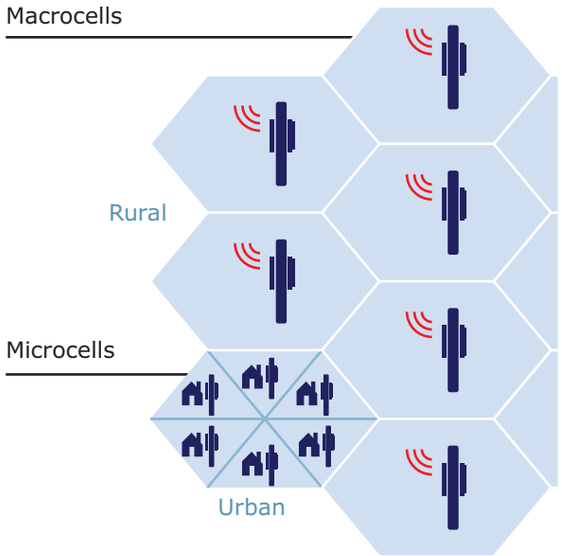


Figure 11.5 Mobile Network

Mobile networking assign to the technology supports voice/data, network connectivity using via radio transmission solution. The common application of mobile networks is mobile phones, tablets, etc.. In the past, wireless communications largely used circuit switching to carry only voice over a network, but now currently both data and voice are being transmitted over both circuit via switched networks and packet-switched networks.

The generation of mobile networks are as follows.

- First Generation(1G) 1981- NMT launch
- Second Generation(2G) 1991-GSM Launch
- Second to Third Generation Bridge (2.5)2000 – GPRS launch
- Third Generation(3G) 2003- UK 3G launch
- Fourth Generation (4 G) 2007
- Fifth Generation (5G) 2019+

First Generation (1G) 1981 – NMT launch

During the initial periods the mobile systems were based on analog transmission. NMT stands for Nordic Mobile Telephone communication. They had a very low traffic density of one call per radio channel, and a very poor voice quality, and they used unsure and unencrypted transmission, which leads to the spoofing of its identities.

Second Generation (2G) 1991 – GSM launch

Later the second generation of mobile systems were placed on digital transmission with GSM. **GSM stands for (Global System for Mobile communication)** was most popular standard which is used in second generation, using 900MHz and 1800MHz for the frequency bands. GSM mobile systems grown digital transmission using SIM. SIM stands for (**Subscriber Identity Module)** technology to authenticate a user for identification and billing purposes, and to encrypt the data to prevent listen without permission (eavesdropping). The transmission used as TDMA. TMDA stands for (**Time Division Multiple Access**) and CDMA stands for **(Code Division Multiple Access)** method to increase the amount of information transported on the network. Mobility is supported at layer 2, which stops seamless roaming across assorted access networks and routing domains. This means that

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